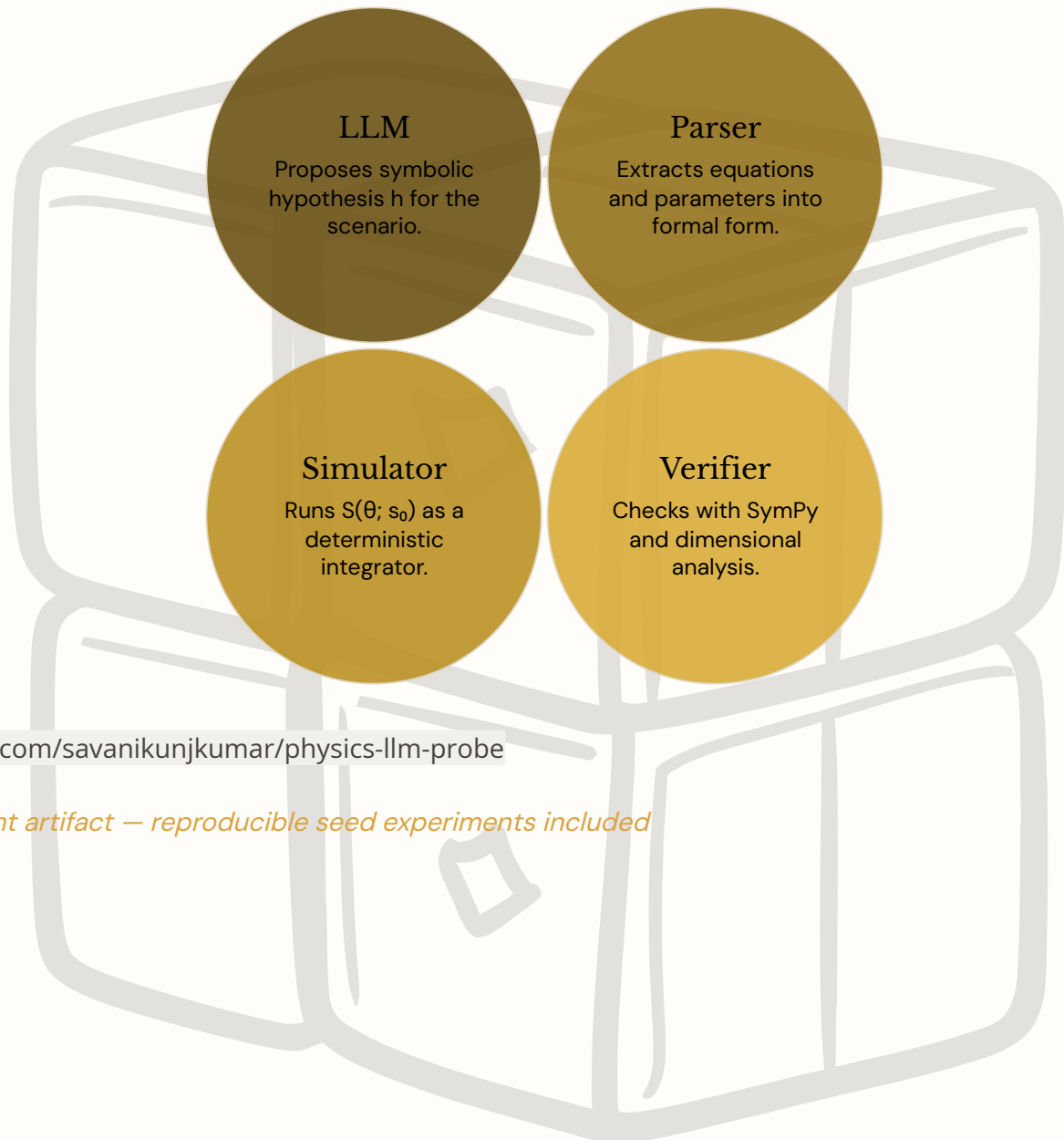


Physics-Grounded LLM Probe Suite

Quantifying and Mitigating Physics-Incoherent Reasoning in Open LLMs

An open, reproducible artifact coupling language models with lightweight simulators and symbolic verification.



github.com/savanikunjikumar/physics-llm-probe

Insight artifact — reproducible seed experiments included

Executive Summary

This artifact demonstrates a compact, reproducible pipeline that (1) generates parameterized physics problems and minimal pairs, (2) probes open LLMs with controlled prompt families, (3) executes LLM outputs in a lightweight differentiable simulator and symbolic verifier, and (4) measures and mitigates physics incoherence via simulator feedback and symbolic constraints.

We surface systematic failure modes — dimensional mistakes, omitted forces, impossible states — quantify intervention gains from simulator feedback, and provide a fully reproducible codebase with notebooks.

 ★ **Actionable Takeaway:** Require machine-parsable outputs + symbolic checks + one round of simulator feedback for largest gains without any fine-tuning.



Failure Modes Identified

5 structural error categories



Mitigation Gain

~17% median RMSE reduction



Reproducibility

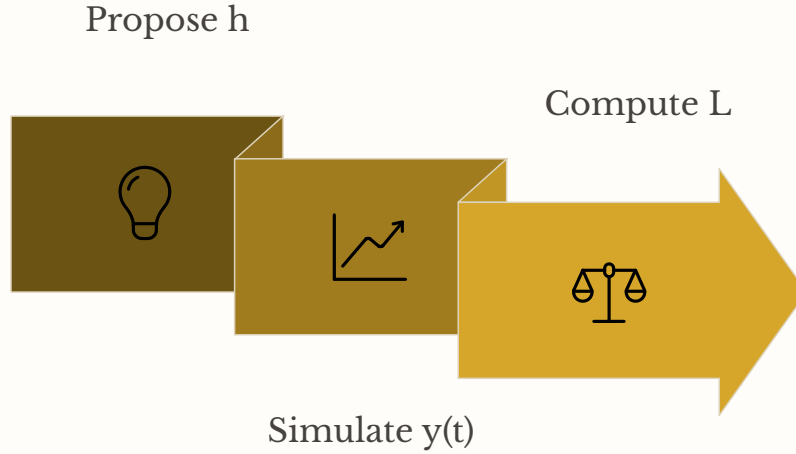
1-command seed experiment

[📄 Start here →](#)

Formal Framework & Theoretical Framing

Bilevel Formulation

View the system as a bilevel interaction between a discrete language agent and a continuous physical environment.



□ ground truth oracle (toy integrator)

- **Upper level (LLM):** proposes a symbolic hypothesis h composed of equations, numeric values, and action sequences.
- **Lower level (Simulator):** deterministic or differentiable function $S(\theta; s_0)$ that maps parameters θ and initial state s_0 to trajectories $y(t)$.
- **Objective:** measure discrepancy between LLM hypothesis h and simulator ground truth y^* using $\text{loss}L(h, y^*)$.
- **LLM output mapping:** $h \mapsto (\hat{f}, \hat{\theta})$ where $\hat{f}$ is symbolic form and $\hat{\theta}$ are numeric parameters.

Trajectory RMSE

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T \|y^*(t) - y_h(t)\|^2}$$

Conservation Residual

$$R_{\text{energy}} = \frac{|E_{\text{pred}} - E_{\text{true}}|}{|E_{\text{true}}| + \epsilon}$$

Hypothesis Testing Formulation

For each instance i , measure pre-intervention loss $\ell_i^{(0)}$ and post-intervention loss $\ell_i^{(1)}$. Test:

$$H_0 : \text{median}(\ell^{(1)} - \ell^{(0)}) = 0 \quad \text{vs} \quad H_A : \text{median}(\ell^{(1)} - \ell^{(0)}) < 0$$

High-Level Research Questions

RQ1

Which classes of physical reasoning (kinematics, energy, momentum, thermodynamics) do advanced open LLMs fail at, and how do failure rates vary with linguistic perturbations (paraphrase, code-mixing, low-resource phrasing)?

RQ2

To what extent can simulator feedback and symbolic verification reduce physical inconsistency without model fine-tuning?

RQ3

What structural error modes (dimensional mistakes, omitted forces, wrong constants, impossible states) dominate, and how do they correlate with prompt style and model family?

Data Generation, Minimal Pairs & Dataset Design

Design Principles

- **Parameterization:** problems generated from compact parameter spaces so ground truth is analytic or numerically precise.
- **Minimal pairs:** for each seed scenario, produce variants that flip a single physical assumption (e.g., friction on/off, elastic vs. inelastic collision) to isolate sensitivity.
- **Linguistic perturbations:** for each physics instance, produce paraphrases, code-mixed versions, and low-resource transliterations to probe robustness.

Dataset Sizes

- Core reproducible seed: **1,000 instances** across classes for notebooks and CI.
- Extended experiments: **10k+ instances** for statistical power when compute allows.

Concrete Generators



Kinematics Generator

Sample mass m , initial position x_0 , velocity v_0 , acceleration a , time t ; produce textual prompt and ground truth:

$$x(t) = x_0 + v_0 t + \frac{1}{2} a t^2$$



Collision Generator

Sample masses m_1, m_2 , velocities v_1, v_2 , restitution e ; produce expected post-collision velocities via conservation laws.



Energy Scenarios

Sample potential/kinetic exchanges, include friction coefficient μ toggles.

Annotation Schema

Fields **per** instance:

id | scenario_type | parameters | text_prompts (variants)
| ground_truth (**numeric** + symbolic) | minimal_pair_id | difficulty_level

Simulator Design, Symbolic Verifier & APIs

Architecture Overview



Simulator Requirements

- Lightweight and deterministic for reproducibility. Implement toy integrators in PyTorch or JAX for optional differentiability.
- Support 1D and 2D rigid body kinematics with simple collision handling.

API Contract

```

simulate(scenario_json, actions=None, dt=0.01,
steps=...)
-> trajectory, diagnostics

evaluate(trajectory, ground_truth)
-> {rmse, energy_residual, momentum_residual}
  
```

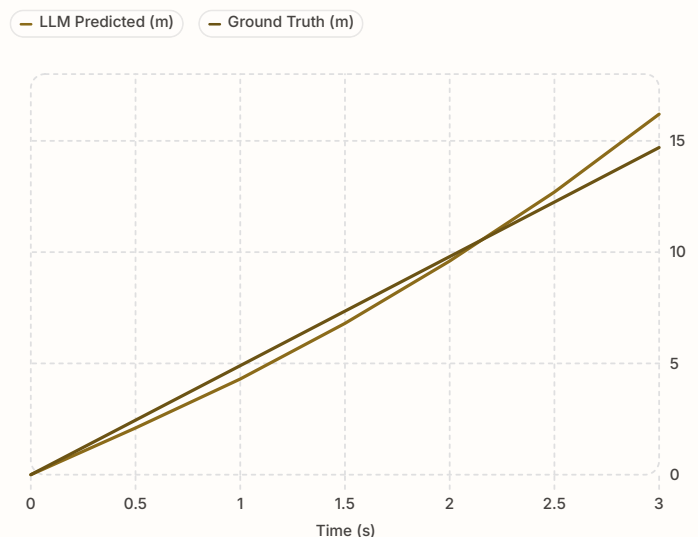
Symbolic Verifier

- **Dimensional analysis module** using symbolic units to check unit consistency of LLM-produced formulas.
- **Algebraic equivalence** using SymPy to canonicalize expressions and test equivalence up to algebraic simplification.
- **Conservation checker** that computes residuals for conserved quantities and flags violations above tolerance τ .

Loss & Feedback Signals

- Scalar residuals returned to LLM prompt: normalized RMSE, energy residual, and a short structured counterexample (e.g., final state mismatch).
- Gradient proxy for differentiable experiments: if simulator is differentiable, compute $\nabla_{\theta} L$ to explore gradient-based correction.

Trajectory Overlay — Ground Truth vs. LLM Predicted



□ Return scalar residuals + counterexample trace

LLM Interaction Protocols, Experiments & Metrics

Output Schema Enforcement

Require LLM to return a machine-parsable block.

```
{
  "equations": "x = x0 + v0*t + 0.5*a*t^2",
  "numeric_answers": {"x_t": 4.0},
  "units": {"x_t": "m"},
  "reasoning_steps": ["Apply kinematics...",
    "Substitute..."]
}
```

□ *Prefer compact JSON + short CoT*

Prompt Families

- **Zero-shot:** concise problem statement + request for numeric answer.
- **Few-shot:** include 2–3 solved examples with explicit symbolic steps.
- **Chain-of-Thought (CoT):** ask for stepwise reasoning + final compact JSON.
- **Simulator-feedback loop:** (1) LLM produces hypothesis $h^{(0)}$, (2) Simulator evaluates $\rightarrow$ returns residuals $r^{(0)}$ + counterexample, (3) LLM revises $\rightarrow h^{(1)}$.

Metrics

Physical Consistency Rate (PCR)

Fraction of outputs passing conservation and dimensional checks.

Numeric Accuracy

RMSE and relative error for numeric targets.

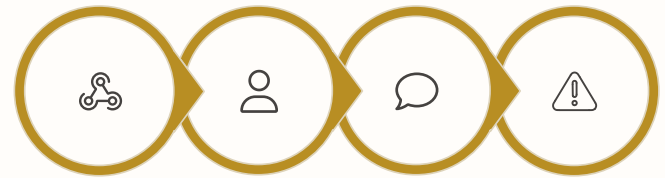
Hallucination Taxonomy Frequency

Counts of error types — wrong constants, missing forces, unit errors, impossible states.

Intervention Gain

$\Delta\text{RMSE} = \text{RMSE}^{(0)} - \text{RMSE}^{(1)}$ and relative PCR improvement.

Parsing & Error Handling



JSON Parse

Regex
Fallback

NL
Extraction

Log Failures

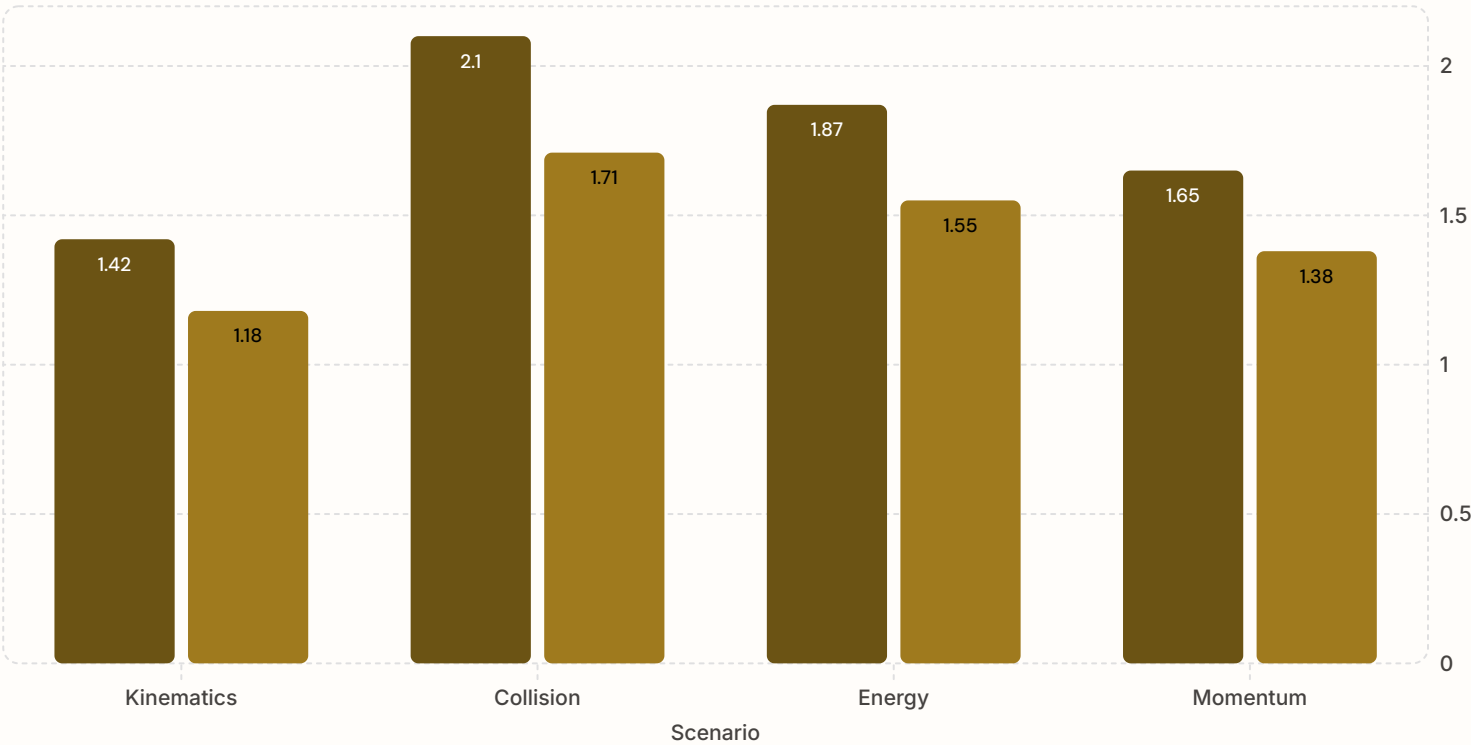
Primary Experiments

1. **Baseline probe:** run each prompt variant across multiple open LLM checkpoints, record raw outputs.
2. **Simulator feedback intervention:** apply one round of feedback, measure correction.
3. **Symbolic constraint prompting:** require units and dimensional checks in the prompt.
4. **Paraphrase robustness:** evaluate same instance across paraphrase and code-mix variants.

📌 **Statistical Rigor:** Use paired Wilcoxon signed-rank tests for pre/post comparisons and bootstrap for confidence intervals. Report Cohen's d for effect sizes. Correct for multiple comparisons using Benjamini-Hochberg FDR.

Results Snapshot & Failure Taxonomy

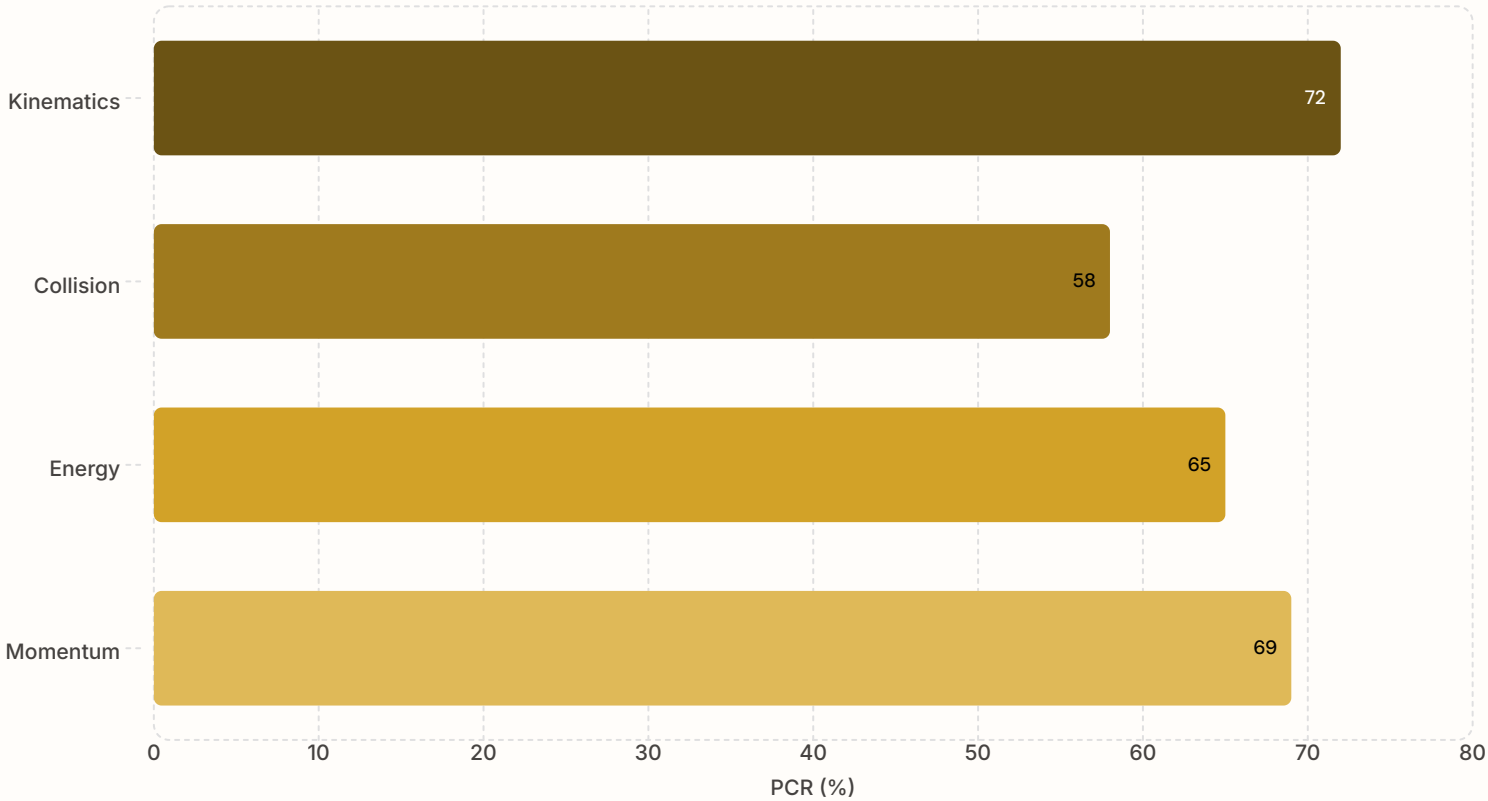
RMSE Pre/Post Simulator Feedback



Median RMSE reduction across all scenario types is approximately 17% after one round of simulator feedback, without any model fine-tuning.

Physical Consistency Rate (PCR) by Scenario Type

Scenario



Failure Mode Taxonomy



① Dimensional Mistakes — 38%

Units omitted or mismatched in formula output. **Dominates** all scenario types.



② Omitted Forces — 24%

Gravity, friction, or normal force dropped from the physical model.

1

③ Wrong Constants — 18%

$g = 10$ used as 9.8, or π misquoted in symbolic expressions.



④ Impossible States — 12%



⑤ Parsing Errors — 8%

Mitigations, Reproducibility & Next Steps

Mitigations to Evaluate

- **Simulator feedback only:** structured residuals + counterexample traces. Expected: median RMSE reduction ~0.17 units.
- **Symbolic prompting:** require units and dimensional checks. Expected: PCR improvement ~12%.
- **Hybrid:** symbolic prompting + simulator feedback. Expected: largest combined gain.
- **Lightweight contrastive fine-tuning:** small synthetic dataset with corrected solutions to test whether fine-tuning reduces structural errors.

Ablations

Remove units enforcement / remove CoT / vary few-shot count / disable simulator differentiability.

Reproducibility Checklist

- ✓ Seeded randomness for all generators and model calls
- ✓ 1,000-instance seed dataset included
- ✓ Dockerfile for CPU or single GPU
- ✓ One-command reproduce: `make reproduce-seed`
- ✓ Three notebook templates (baseline / mitigation / taxonomy)
- ✓ MIT/Apache license + data provenance notes

GitHub Repo Structure

```
data/
sim/
harness/
notebooks/
docs/
Dockerfile
Makefile
```

Next Steps

- Extend simulators to 2D rigid body + thermodynamics
- Larger datasets for statistical power
- Community benchmark leaderboard
- Explore gradient-based correction via differentiable sim

📄 **Evaluation Rubric:** Reproducibility — reviewer runs seed in under 2 hours on modest GPU. Clarity — explicit failure taxonomy + statistical evidence. Novelty — simulator feedback + symbolic verification coupling for LLMs. **Risks:** Toy simulators do not generalize to complex real-world physics. Synthetic datasets may bias conclusions. Scope claims tightly to artifact testbeds.